

# Linear regression (simple and inference)

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# Simple Linear Regression: The Foundations

Statistical Foundations

Inference in Simple Linear Regression



What lifestyle factors do you think are predictive of body fat percentage?



**Problem:** Predict body fat percentage based on observed lifestyle characteristics

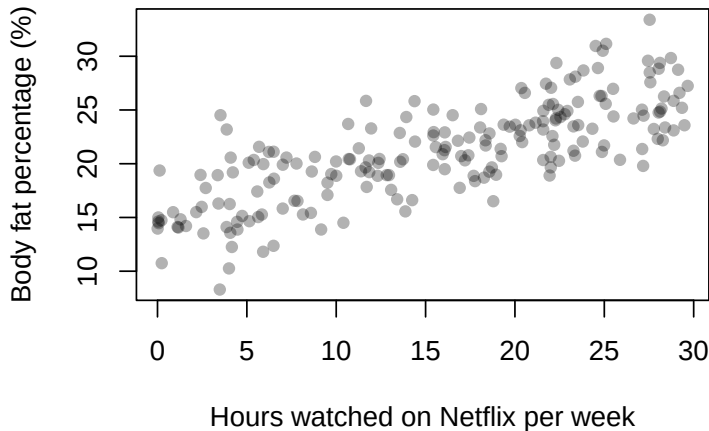
**Solution:**

- Look at data where we know body fat percentage and some observed characteristics
- Build a decision rule that predicts body fat percentage as a function of the observed characteristics

To keep things simple, let's focus only on **hours watched on Netflix per week**.



- The value we seek to predict is the **response** (or dependent/output) variable:
  - $Y$  = body fat percentage (%)
- The variable we use to guide prediction is the **predictor** (or explanatory/input) variable:
  - $X$  = hours watched on Netflix per week



View the data as points in the  $X \times Y$  plane. There seems to be a **linear** relationship!



$Y$  = response or outcome variable

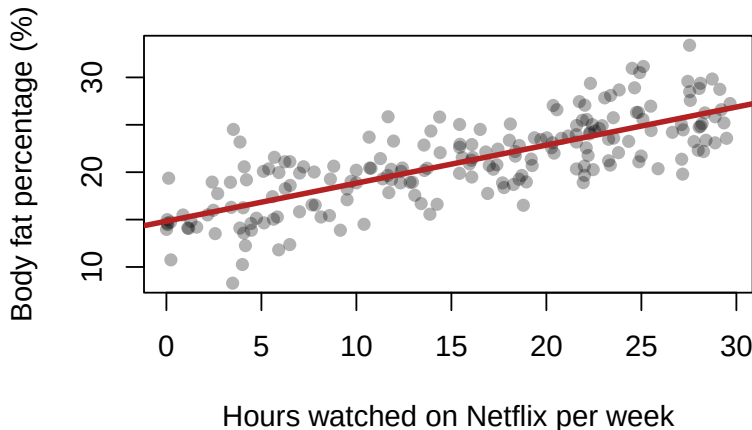
$X$  = explanatory or predictor variable

A **linear** relationship is written:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

- $\beta_0$  is the **intercept** (in units of  $Y$ )
- $\beta_1$  is the **slope** (in units of  $Y$  per unit of  $X$ )
- $\varepsilon$  is the **error** or residual

Recall the equation of a line:  $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$



As Netflix hours go up, body fat percentage tends to go up. But how do we find the “best” line?



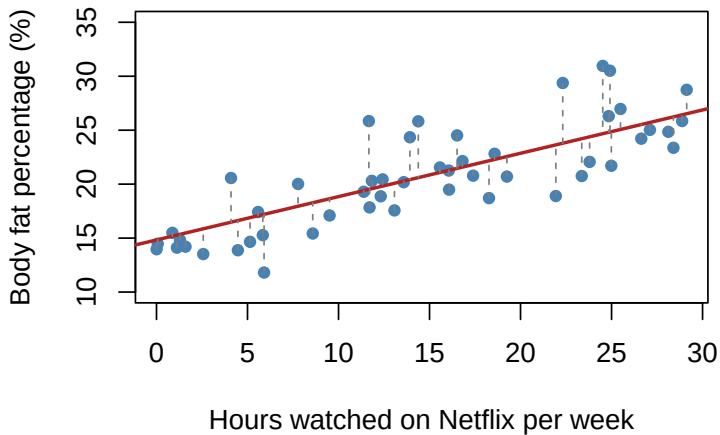
- The **fitted value**  $\hat{Y}_i$  is our prediction for observation  $i$ :

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$$

- The **residual**  $e_i$  is how wrong our prediction is:

$$e_i = Y_i - \hat{Y}_i$$

- We can write:  $Y_i = \hat{Y}_i + e_i$  (observed = predicted + error)



The dashed lines are the residuals:  $e_i = Y_i - \hat{Y}_i$



How do we find the “best” line?

Ideally, we want to minimize the size of all residuals:

- If they were all zero, we'd have a perfect fit
- But there's a trade-off: moving closer to some points means moving away from others



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**Least Squares:** Choose  $\hat{\beta}_0$  and  $\hat{\beta}_1$  to minimize  $\sum_{i=1}^n e_i^2$

$$\sum_{i=1}^n e_i^2 = (Y_1 - \hat{Y}_1)^2 + (Y_2 - \hat{Y}_2)^2 + \cdots + (Y_n - \hat{Y}_n)^2$$



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- Leads to nice mathematical properties (unique solution)
- The formulas for  $\hat{\beta}_0$  and  $\hat{\beta}_1$  are:

$$\hat{\beta}_1 = r_{xy} \times \frac{s_y}{s_x}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

where  $r_{xy}$  is the correlation and  $s_x$ ,  $s_y$  are standard deviations

```
model <- lm(Body_Fat ~ Netflix_Hours, data=netflix_data)
summary(model)
```

```
Call:
lm(formula = Body_Fat ~ Netflix_Hours, data = netflix_data)
```

```
Residuals:
```

| Min     | 1Q      | Median | 3Q     | Max    |
|---------|---------|--------|--------|--------|
| -7.9344 | -1.8695 | 0.0093 | 1.8318 | 8.2710 |

```
Coefficients:
```

|               | Estimate | Std. Error | t value | Pr(> t )   |
|---------------|----------|------------|---------|------------|
| (Intercept)   | 14.82928 | 0.41936    | 35.36   | <2e-16 *** |
| Netflix_Hours | 0.40153  | 0.02338    | 17.18   | <2e-16 *** |

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 2.889 on 198 degrees of freedom
```

```
Multiple R-squared:  0.5984, Adjusted R-squared:  0.5964
```

```
F-statistic: 295 on 1 and 198 DF, p-value: < 2.2e-16
```



Our prediction equation is:

$$\widehat{\text{Body Fat}} = 14.83 + 0.4 \times \text{Netflix Hours}$$

- **Intercept** ( $\hat{\beta}_0 = 14.83$ ): Predicted body fat percentage when Netflix Hours = 0.



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- **Intercept** ( $\hat{\beta}_0 = 14.83$ ): Predicted body fat percentage when Netflix Hours = 0.
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- A 10 hour increase  $\rightarrow$  4 percentage point increase in predicted body fat

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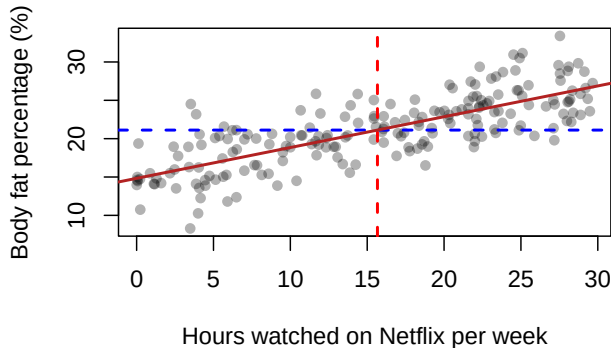


To understand how we get the slope and intercept, we need to understand how  $X$  and  $Y$  vary together.

- **Variance** measures how one variable spreads out:  $s_y^2 = \frac{1}{n-1} \sum (Y_i - \bar{Y})^2$
- **Covariance** measures how two variables vary *together*:

$$\text{cov}(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

- Positive covariance: When  $X$  is above its mean,  $Y$  tends to be above its mean
- Negative covariance: When  $X$  is above its mean,  $Y$  tends to be below its mean
- Zero covariance: No linear relationship



The covariance measures how points fall relative to the mean lines. Points in quadrants I and III contribute positively; points in II and IV contribute negatively.



**Covariance** is hard to interpret because it depends on the units of measurement.

**Correlation** standardizes the covariance:

$$r = \text{corr}(X, Y) = \frac{\text{cov}(X, Y)}{s_x s_y} = \frac{\text{cov}(X, Y)}{\sqrt{s_x^2 s_y^2}}$$



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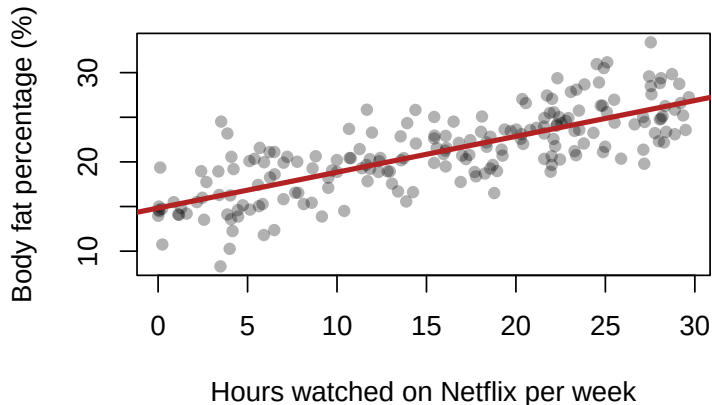
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- $r > 0$ : positive linear relationship
- $r < 0$ : negative linear relationship
- $r = 0$ : no linear relationship
- $|r| = 1$ : perfect linear relationship

## Correlation for our Netflix example



```
cor(netflix_data$Netflix_Hours, netflix_data$Body_Fat)
```

```
[1] 0.7735597
```



How do we interpret this correlation?



Remember the formulas for the slope and intercept:

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- It also depends on the ratio of standard deviations
- If correlation is zero, the slope is zero (no linear relationship)
- The intercept ensures the regression line passes through  $(\bar{X}, \bar{Y})$



Two key questions:

1. How much of the variation in  $Y$  does our model explain?
2. How wrong are our predictions, on average?



Remember:  $Y = \hat{Y} + e$  (observed = predicted + residual)

The total variation in  $Y$  can be split:

$$\underbrace{\sum (Y_i - \bar{Y})^2}_{\text{SST (Total)}} = \underbrace{\sum (\hat{Y}_i - \bar{Y})^2}_{\text{SSR (Regression)}} + \underbrace{\sum e_i^2}_{\text{SSE (Error)}}$$

- **SST**: Total variation in  $Y$
- **SSR**: Variation explained by the regression
- **SSE**: Variation left unexplained (residuals)



$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

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- About 59.8% of the variation in body fat percentage is explained by Netflix hours
- Note: For simple regression,  $R^2 = r^2$  (correlation squared)



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- For our model:  $s_e = 2.89$
- On average, our predictions are off by about 2.89 percentage points
- Why  $n - 2$ ? We “used up” 2 degrees of freedom estimating  $\hat{\beta}_0$  and  $\hat{\beta}_1$



The **simple linear regression model** assumes:

$$Y = \beta_0 + \beta_1 X + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2)$$

- $\beta_0 + \beta_1 X$  is the “true line” (systematic part)



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- This means:  $Y|X \sim N(\beta_0 + \beta_1 X, \sigma^2)$



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*Let's see these ideas in action with a new example!*

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What personal characteristics about an instructor do you think are predictive of the scores they receive on student evaluations?



## Economics of Education Review

Volume 24, Issue 4, August 2005, Pages 369–376

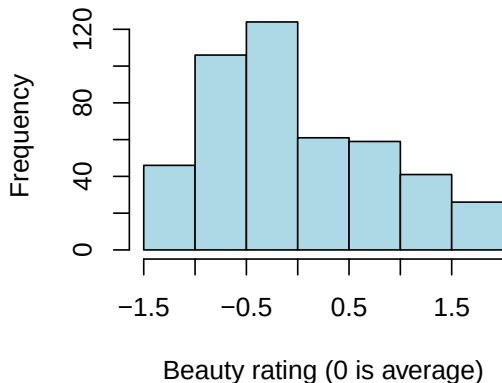
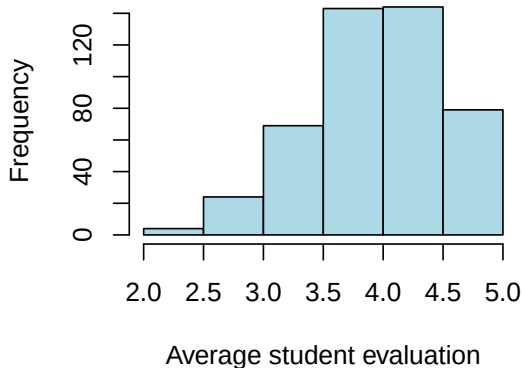


### Beauty in the classroom: instructors' pulchritude and putative pedagogical productivity

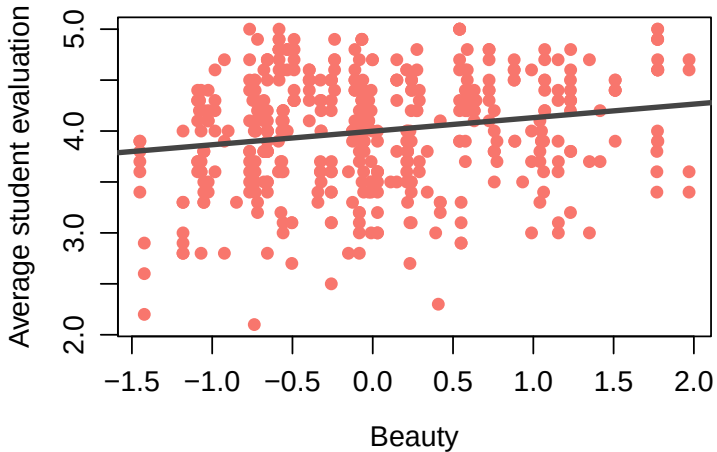
Daniel S. Hamermesh  , Amy Parker



- Student evaluations of  $N = 463$  instructors at UT Austin, 2000-2002
- For each instructor:
  - **eval**: average student evaluation of teacher
  - **beauty**: average beauty score from a six-student panel
  - **gender**: male or female
  - **credits**: single- or multi-credit course
  - **age**: age of instructor
  - (and more...)



Do you think there is a positive or negative relationship between beauty and teaching ratings?



```
cor(profs$eval, profs$beauty)
```

```
[1] 0.189
```

# Let's build a simple regression model



```
model <- lm(eval ~ beauty, data=profs)
summary(model)
```

```
Call:
lm(formula = eval ~ beauty, data = profs)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-1.8002 -0.3630  0.0725  0.4021  1.1037
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.9983     0.0253   157.73 < 2e-16 ***
beauty         0.1330     0.0322    4.13 4.2e-05 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.545 on 461 degrees of freedom
Multiple R-squared:  0.0357, Adjusted R-squared:  0.0336
F-statistic: 17.1 on 1 and 461 DF,  p-value: 4.25e-05
```



- `eval` is the **response** variable ( $Y$ ); `beauty` is the **predictor** variable ( $X$ )



- eval is the **response** variable ( $Y$ ); beauty is the **predictor** variable ( $X$ )
- Simple regression uses the best fit line to give us a linear equation to predict  $Y$  from  $X$ :

$$\widehat{\text{eval}} = 3.998 + 0.133 \cdot \text{beauty}$$



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- We can predict the evaluation score for someone based on their beauty score. What is the predicted evaluation score for someone with a beauty score of 1?

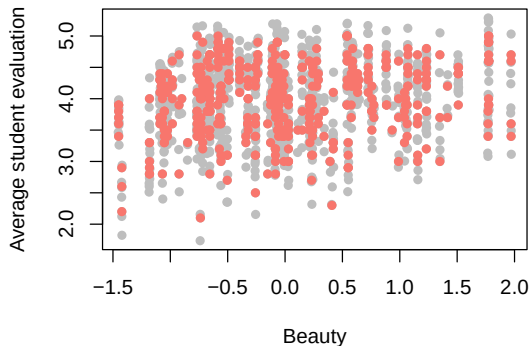


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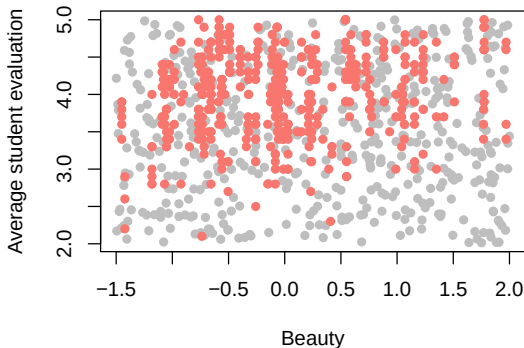
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- What do the coefficients mean?

Is this the population?

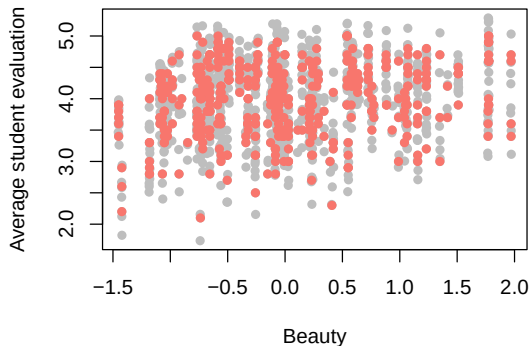


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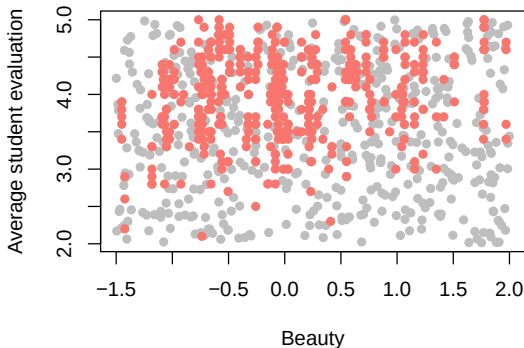


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Or is this the population?



- The population line  $Y = \beta_0 + \beta_1 X$  is what we actually care about, and we can never see it
- $\hat{\beta}_0$  and  $\hat{\beta}_1$  are our best estimates of it. Inference is about how close they are.



|             | Estimate | Std. Error | t value | Pr(> t )    |
|-------------|----------|------------|---------|-------------|
| (Intercept) | 3.9983   | 0.0253     | 157.73  | < 2e-16 *** |
| beauty      | 0.1330   | 0.0322     | 4.13    | 4.2e-05 *** |

- 95% CI for the slope is  $\hat{\beta}_1 \pm t^*SE(\beta_1)$  (estimate  $\pm$  (critical value)(standard error))



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- $t^*$  is the critical value from a  $t$  distribution with  $n - 2$  degrees of freedom:  $qt(0.975, 463-2) \approx 1.97$
- So the 95% CI for the slope is  $0.133 \pm (1.97)(0.0322)$
- **Rule of thumb** for anything in this class: (estimate)  $\pm 2$ (standard errors)



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|-------------|----------|------------|---------|-------------|
| (Intercept) | 3.9983   | 0.0253     | 157.73  | < 2e-16 *** |
| beauty      | 0.1330   | 0.0322     | 4.13    | 4.2e-05 *** |

- 95% CI for the slope is  $\hat{\beta}_1 \pm t^*SE(\beta_1)$  (estimate  $\pm$  (critical value)(standard error))
- $t^*$  is the critical value from a  $t$  distribution with  $n - 2$  degrees of freedom:  $qt(0.975, 463-2) \approx 1.97$
- So the 95% CI for the slope is  $0.133 \pm (1.97)(0.0322)$
- **Rule of thumb** for anything in this class: (estimate)  $\pm 2$ (standard errors)
- The standard error measures how far our *estimate* is likely to be from the *population value*

```
confint(model)
```

|             | 2.5 %  | 97.5 % |
|-------------|--------|--------|
| (Intercept) | 3.9485 | 4.048  |
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- $\beta_1$ : We are 95% confident that the incremental impact of each additional beauty point is between 0.07 and 0.20 student evaluation points.



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- $\beta_1$ : We are 95% confident that the incremental impact of each additional beauty point is between 0.07 and 0.20 student evaluation points.
- $\beta_0$ : We are 95% confident that the average student evaluation score for average-looking professors (beauty = 0) is between 3.95 and 4.05.



95% confidence intervals for parameters (slope, intercept, etc): The interval contains the true value in 95% of all possible datasets drawn from the population.



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95% confidence intervals for predictions (aka prediction intervals): The interval captures 95% of *future  $y$  values* for a given  $x$

## Testing whether the slope is really zero



|        | Estimate | Std. Error | t value | Pr(> t )    |
|--------|----------|------------|---------|-------------|
| beauty | 0.1330   | 0.0322     | 4.13    | 4.2e-05 *** |

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- Test the null  $\beta_1 = 0$  against the alternative  $\beta_1 \neq 0$ , accepting a false positive rate (type I error) of  $\alpha = 0.05$
- $\alpha = 0.05$  means we accept a 5% chance of rejecting a true null. You must pick it before looking at the data.



|        | Estimate | Std. Error | t value | Pr(> t )    |
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| beauty | 0.1330   | 0.0322     | 4.13    | 4.2e-05 *** |

- The  $p$ -value for the slope is  $4.2 \times 10^{-5}$ ; since it is very small, it is very unlikely that  $\hat{\beta}_1$  is this big just by chance if  $\beta_1$  is truly zero



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- Connection to CIs: a 95% CI contains *all the values* the test fails to reject at  $\alpha = 0.05$ , i.e. a **set of plausible values** your data doesn't rule out

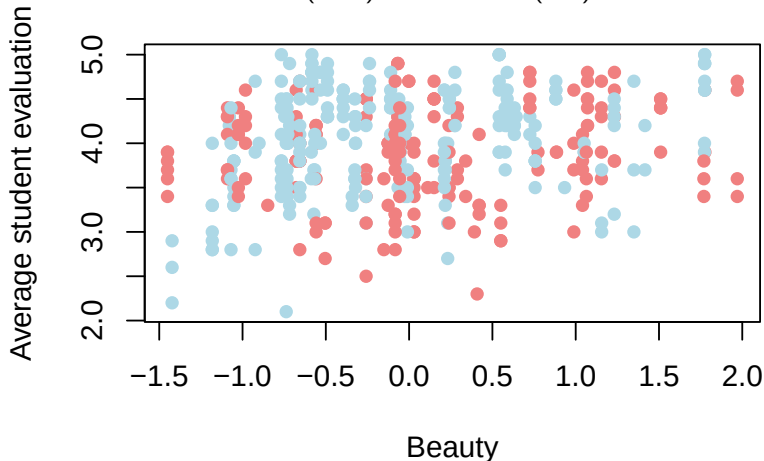
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(Spoiler alert: No.)

# What's the impact of gender on student evaluations?



Do you see a difference between men (blue) and women (red)?



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